Contexts and pseudocontexts

Mirko Navara
Faculty of Electrical Engineering, Czech Technical University in Prague, Technická 2, CZ-166 27 Prague 6, Czech Republic[*]

Karl Svozil
Institute for Theoretical Physics, TU Wien, Wiedner Hauptstrasse 8-10/136, A-1040 Vienna, Austria[†]

(Dated: April 23, 2026)

In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis—a *context*—is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator, placing the notion naturally among finite frame-theoretic constructions. Size $k=1$ is trivial; size $k=2$ already admits genuinely nonorthogonal examples over $\mathbb{C}$; and, in fact, examples exist for every $k \geq 2$. For $k \geq 3$ genuine pseudocontexts also exist over $\mathbb{R}$. We present the operator formulation, give a general explicit construction, analyse the spectral structure of the associated structure operator, illustrate the symmetric $k=3$ case with a figure, and list the remaining classification problems.

Keywords: projection lattice, pseudocontext, rank-one decomposition, structure operator, quantum contextuality

I. INTRODUCTION

Let L be the lattice of projections in a Hilbert space of finite dimension $n \geq 3$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P\colon L \to [0,1]$. We denote by $S(L)$ the set of all states on L.

Vectors $a_1, \ldots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case) they satisfy

$$\forall P \in S(L): \quad \sum_{i=1}^{n} P(a_i) = 1. \tag{1}$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [1] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. The answer is simple: pseudocontexts already exist for $k=2$ over $\mathbb{C}$, and, more generally, for every $k \geq 2$; for $k \geq 3$ they exist over $\mathbb{R}$ as well.

II. PRELIMINARIES

Throughout, H is a complex Hilbert space with $\dim H = n \geq 3$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector a and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans.

A *state* is a map $P\colon L \to [0,1]$ with $P(I)=1$ that is finitely additive on orthogonal families. By Gleason's theorem [2], every state arises from a density operator ρ via

$$P(a) = \operatorname{tr}(\rho P_a), \qquad \rho \geq 0, \quad \operatorname{tr}\rho = 1. \tag{2}$$

The key consequence we use repeatedly is that two self-adjoint operators that agree on every state are equal as operators.

III. CONTEXTS AND PSEUDOCONTEXTS

The operator form of (1) is elementary and is the starting point for everything that follows.

Proposition 1 (Contexts)**.** *Unit vectors $a_1, \ldots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^{n} P_{a_i} = I. \tag{3}$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j:

$$1 = \langle a_j, I a_j \rangle = \sum_{i=1}^{n} |\langle a_i, a_j \rangle|^2.$$

The term with $i=j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j \rangle = 0$ for $i \neq j$. As this holds for every j, the family is orthonormal, and being of size n in an n-dimensional space it is an orthonormal basis. ☐

Combined with (2), Proposition 1 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 2 (Pseudocontext)**.** Let $k \geq 1$. A *pseudocontext of size k* is a pair of k-tuples of atoms $(a_1, \ldots, a_k)$, $(b_1, \ldots, b_k)$ such that

[*] navara@fel.cvut.cz
[†] karl.svozil@tuwien.ac.at

(i) each tuple contains pairwise distinct atoms,

(ii) the two tuples are disjoint as sets of rays, so that $|\{a_1,\dots,a_k,b_1,\dots,b_k\}| = 2k$, and

(iii) $\forall P \in S(L): \quad \sum_{i=1}^{k} P(a_i) = \sum_{i=1}^{k} P(b_i).$

A pseudocontext is called *genuine* if neither k-tuple is a context.

Remark 3. Condition (i) excludes repeated atoms within a single tuple; condition (ii) ensures the two tuples are truly distinct as collections of rays. Both conditions are part of the definition; without them the notion degenerates trivially.

By (2), condition (iii) of Definition 2 is equivalent to the operator identity

$$T := \sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}, \tag{4}$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are rank-one decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support one obtains a POVM.

Remark 4 (Connection to SIC-POVMs). When the common structure operator satisfies $T = \frac{k}{n} I$ (tight frame on all of H), each family $\{P_{a_i}\}$ and $\{P_{b_i}\}$ scaled by n/k forms a symmetric informationally complete POVM (SIC-POVM) if $k = n^2$ and the frame vectors are equiangular [3]. Pseudocontexts therefore generalise the SIC-POVM notion by dropping both the tightness and equiangularity requirements.

IV. THE SMALLEST CASES

Proposition 5 (Size $k = 1$)**.** *There is no pseudocontext of size $k = 1$. This holds for every $n \geq 1$.*

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ is equivalent, via (2), to $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting the distinctness condition (ii) of Definition 2. □

Remark 6 (Minimal ambient dimension). Every pseudocontext of size $k \geq 2$ can be realised inside a two-dimensional subspace, and hence embedded into $\mathbb{C}^n$ for any $n \geq 2$: all constructions in Sections IV A and VI live in $\mathrm{span}\{e_1, e_2\}$. For $k \geq 3$ the construction also works over $\mathbb{R}^n$. No pseudocontext of any size exists in dimension $n = 1$, since there every unit vector spans the unique ray, making the distinctness condition $|\{a_i\} \cup \{b_i\}| = 2k$ impossible.

A. Pseudocontexts of size $k = 2$

The $k = 2$ case requires complex scalars; we show below that no genuine real pseudocontext of size $k = 2$ exists, while the complex case allows such configurations.

Corollary 7 (Size $k = 2$, complex)**.** *There are genuinely nonorthogonal pseudocontexts of size $k = 2$ in $\mathbb{C}^n$ for every $n \geq 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors and set

$$u_\pm = \left(\tfrac{\sqrt{3}}{2}, \pm\tfrac{1}{2}, 0, \dots, 0\right), \qquad v_\pm = \left(\tfrac{\sqrt{3}}{2}, \pm\tfrac{i}{2}, 0, \dots, 0\right).$$

A direct computation gives

$$P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}.$$

The four rays are distinct: u_+ and u_- differ by a sign in the real second coordinate; v_+ and v_- differ by a sign in the imaginary second coordinate; and $u_\pm$ versus $v_\pm$ differ by a factor of i in the second coordinate, which is not a real scalar. Within each pair, $\langle u_+, u_- \rangle = \frac{1}{2} \neq 0$ and $\langle v_+, v_- \rangle = \frac{1}{2} \neq 0$, so neither pair is orthogonal, and hence neither pair is a context. The pseudocontext is therefore genuine. □

V. COMPLETE CHARACTERISATION IN DIMENSION $d = 2$

In dimension two the pseudocontext condition admits a complete characterisation in terms of Bloch vectors. This yields a unified and basis-independent description of all pseudocontexts and clarifies the real–complex dichotomy.

A. Bloch representation

Every rank-one projector on $\mathbb{C}^2$ can be written as

$$P_a = \tfrac{1}{2}\left(I + \vec{r}_a \cdot \vec{\sigma}\right), \qquad \vec{r}_a \in \mathbb{R}^3, \quad \|\vec{r}_a\| = 1, \tag{5}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices.

Proposition 8 (Bloch representation of pseudocontexts in $d = 2$)**.** *Let $(a_1,\dots,a_k)$ and $(b_1,\dots,b_k)$ be k-tuples of unit vectors in $\mathbb{C}^2$ with Bloch vectors $\vec{r}_{a_i}, \vec{r}_{b_i} \in \mathbb{R}^3$. Then the following are equivalent:*

1. *(a_i) and (b_i) form a pseudocontext;*

2. $\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k} P_{b_i}$*;*

3. $\sum_{i=1}^{k} \vec{r}_{a_i} = \sum_{i=1}^{k} \vec{r}_{b_i}$.

The common structure operator T has Bloch form

$$T = \tfrac{k}{2}I + \tfrac{1}{2}\vec{R}\cdot\vec{\sigma}, \qquad \vec{R} = \sum_{i=1}^{k}\vec{r}_{a_i} = \sum_{i=1}^{k}\vec{r}_{b_i}.$$

Proof. By Definition 2 and (4), conditions (i) and (ii) are equivalent. For the equivalence with (iii), substitute the Bloch representation (5) into the structure operator:

$$\sum_{i=1}^{k} P_{a_i} = \sum_{i=1}^{k}\tfrac{1}{2}\big(I + \vec{r}_{a_i}\cdot\vec{\sigma}\big) = \tfrac{k}{2}I + \tfrac{1}{2}\Big(\sum_{i=1}^{k}\vec{r}_{a_i}\Big)\cdot\vec{\sigma}.$$

The identity component $\frac{k}{2}I$ is the same for both tuples, so equality of the two structure operators reduces to equality of the vector sums, giving the stated Bloch form of T. □

B. Classification in $\mathbb{C}^2$

Pseudocontexts in dimension two correspond exactly to distinct decompositions of a fixed vector $\vec{R}\in\mathbb{R}^3$ into k unit vectors.

Theorem 9 (Complete characterisation in $\mathbb{C}^2$)**.** *Let $k \geq 1$.*

(i) For $k = 1$, no pseudocontext exists.

(ii) For $k = 2$, pseudocontexts exist and form continuous families.

(iii) For all $k \geq 3$, pseudocontexts exist; if $\vec{R} = 0$, they arise from zero-centroid configurations (e.g. regular polygons).

Proof. Part (i) is Proposition 5.

For (ii), fix $\vec{R} \neq 0$ in $\mathbb{R}^3$ and choose any two distinct decompositions of $\vec{R}$ into two unit vectors, e.g. $\vec{r}_1 + \vec{r}_2 = \vec{R}$ and $\vec{r}_1{}' + \vec{r}_2{}' = \vec{R}$ with $(\vec{r}_1, \vec{r}_2) \neq (\vec{r}_1{}', \vec{r}_2{}')$ as ordered pairs. Such distinct pairs exist because for $\vec{R} \neq 0$ the decomposition is not unique (rotate the pair of vectors about the $\vec{R}$ axis to obtain another distinct pair). By Proposition 8, the corresponding two-tuples form a pseudocontext.

For (iii), take $\vec{R} = 0$. In $\mathbb{R}^3$ there are infinitely many distinct triples of unit vectors with zero sum, e.g. the vertices of any regular triangle centered at the origin, or any rotated version thereof. By Proposition 8, any two such distinct triples yield a pseudocontext of size $k = 3$. □

C. Real–complex dichotomy

Theorem 10 (Real vs. complex in $d = 2$)**.** *(i) In $\mathbb{C}^2$, genuine pseudocontexts of size $k = 2$ exist.*

(ii) In $\mathbb{R}^2$, no genuine pseudocontext of size $k = 2$ exists.

(iii) In $\mathbb{R}^2$, genuine pseudocontexts exist for all $k \geq 3$.

Proof. In $\mathbb{C}^2$, rotations about a fixed Bloch vector $\vec{R} \neq 0$ form a continuous $U(1)$ symmetry, yielding inequivalent decompositions.

In $\mathbb{R}^2$, Bloch vectors lie on a circle. For $k = 2$ the condition $\vec{r}_1 + \vec{r}_2 = \vec{R}$ uniquely determines the pair up to permutation, so no second decomposition exists. If $\vec{R} = 0$, the vectors are orthogonal and form a context.

For $k \geq 3$, distinct planar configurations with identical vector sum exist (e.g. rotated regular polygons). □

D. Spectral form

Proposition 11 (Spectrum in $d = 2$)**.** *The eigenvalues of T are*

$$\lambda_{\pm} = \tfrac{k}{2} \pm \tfrac{1}{2}\|\vec{R}\|.$$

Remark 12. The vector $\vec{R}$ completely parametrises pseudocontexts in dimension two.

Remark 13 (Real vs. complex for $k = 2$). The nonexistence of genuine real pseudocontexts for $k = 2$ in dimension two follows from the classification in Sec. V. Over $\mathbb{R}$, a decomposition of a fixed structure operator into two rank-one terms is unique up to permutation, whereas over $\mathbb{C}$ a continuous phase degree of freedom produces inequivalent decompositions.

This observation was made in [4], without noticing that the situation is different in complex spaces.

VI. PSEUDOCONTEXTS OF EVERY SIZE $k \geq 2$

Theorem 14 (Complex-phase construction)**.** *Let H contain an orthonormal pair e_1, e_2. Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a kth root of unity. Define unit vectors*

$$\begin{aligned} a_j &= \cos\alpha\, e_1 + \sin\alpha\, \omega^j e_2, \\ b_j &= \cos\alpha\, e_1 + \sin\alpha\, e^{i\phi}\omega^j e_2, \qquad j = 0, \ldots, k-1. \end{aligned}$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(\cos^2\alpha\, P_{e_1} + \sin^2\alpha\, P_{e_2}\big).$$

Hence $(a_0, \ldots, a_{k-1})$ and $(b_0, \ldots, b_{k-1})$ form a pseudocontext of size k. If $\alpha \neq \pi/4$, then the pseudocontext is genuine.

Proof. Write $c = \cos\alpha$ and $s = \sin\alpha$. For each j,

$$\begin{aligned} P_{a_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs\big(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j|e_2\rangle\langle e_1|\big), \\ P_{b_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs\big(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j|e_2\rangle\langle e_1|\big). \end{aligned}$$

Summing over $j = 0, \ldots, k-1$ and using $\sum_{j=0}^{k-1}\omega^j = 0$ cancels all off-diagonal terms, giving

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k\big(c^2 P_{e_1} + s^2 P_{e_2}\big).$$

Distinctness. If $a_j = \lambda a_\ell$ for some scalar λ, then comparing first coordinates (which equal $c \neq 0$) gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Hence the a_j are pairwise distinct rays; the identical argument applies to the b_j. If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi}\omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, contradicting the assumption that $e^{i\phi}$ is not a kth root of unity. Hence no a_j coincides with any b_ℓ.

Genuineness. For $j \neq \ell$ within the a-tuple,

$$\langle a_j, a_\ell \rangle = c^2 + s^2 \omega^{\ell - j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. So for $\alpha \neq \pi/4$ no two distinct vectors in the a-tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell \rangle = c^2 + s^2\omega^{\ell-j}$, so the same conclusion holds for the b-tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. □

Remark 15. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2} I_{\mathrm{span}\{e_1,e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k = 2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 17). In dimension $d = 2$, this construction reduces to the Bloch-vector description of Sec. V.

VII. A SYMMETRIC $k = 3$ EXAMPLE

The case $k = 3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes–Benz frame.

Example 16 (Mercedes–Benz type pseudocontext)**.** Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2), and fix an angle θ satisfying $\theta \not\equiv 0$, $\theta \not\equiv \pi/3$, and $\theta \not\equiv 2\pi/3$ modulo π. Put, for $j = 0, 1, 2$,

$$a_j = \cos\!\left(\tfrac{2\pi j}{3}\right) e_1 + \sin\!\left(\tfrac{2\pi j}{3}\right) e_2,$$

$$b_j = \cos\!\left(\tfrac{2\pi j}{3} + \theta\right) e_1 + \sin\!\left(\tfrac{2\pi j}{3} + \theta\right) e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^{2} P_{a_j} = \sum_{j=0}^{2} P_{b_j} = \tfrac{3}{2} I_V.$$

Thus $\{a_0, a_1, a_2\}$ and $\{b_0, b_1, b_2\}$ form a genuine pseudocontext of size $k = 3$ over $\mathbb{R}$.

Proof. In the basis (e_1, e_2), each projector has the form

$$P_{\cos\varphi\, e_1 + \sin\varphi\, e_2} = \begin{pmatrix} \cos^2\varphi & \cos\varphi\sin\varphi \\ \cos\varphi\sin\varphi & \sin^2\varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^{2} \cos(2\varphi_j) = 0, \qquad \sum_{j=0}^{2} \sin(2\varphi_j) = 0,$$

which hold for any arithmetic progression of three angles spaced $2\pi/3$ apart, imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2} I_V$.

The three excluded residue classes modulo π account for all collisions between the two triples:

- $\theta \equiv 0 \pmod{\pi}$: every b_j coincides with a_j or $-a_j$, giving the same ray.
- $\theta \equiv \pi/3 \pmod{\pi}$: the b-triple is a cyclic permutation of the a-triple as a set of rays.
- $\theta \equiv 2\pi/3 \pmod{\pi}$: the b-triple is again a permutation of the a-triple.

For all other values of θ the six rays are pairwise distinct. Genuineness holds because for generic θ no two vectors within the same triple are orthogonal (the angle between consecutive vectors in each triple is $2\pi/3 \neq \pi/2$), so neither triple is a context. □

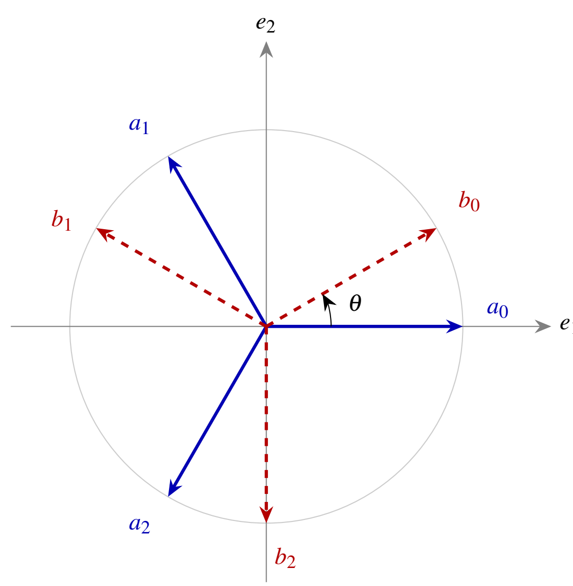

FIG. 1. The Mercedes–Benz pseudocontext of Example 16, drawn for $\theta = \pi/6$. The solid blue rays a_0, a_1, a_2 and the dashed red rays b_0, b_1, b_2 are two distinct unordered triples in the plane V, yet the associated rank-one projectors have the same sum $\frac{3}{2} I_V$.

Remark 17 (Complex trine description). Example 16 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\mathrm{span}\{e_1, e_2\}$. The *trines*

$$u_j = \tfrac{1}{\sqrt{2}}\big(1, \omega^j, 0, \ldots, 0\big), \quad v_j = \tfrac{1}{\sqrt{2}}\big(1, e^{i\phi}\omega^j, 0, \ldots, 0\big), \quad j = 0, 1, 2,$$

yield the same phenomenon after a change of basis in V. The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ cancel, giving the common structure operator $\frac{3}{2} I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

Remark 18 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three; see, for instance, the coordinatisations discussed in [5, 6]. The main difference is that the pseudocontext in Example 16 has only one constant value in the probability sum, while the advanced constructions admit a larger range of values, still respecting the equality. Such examples are useful when one wants to move beyond purely symmetric frame orbits.

Similar constructions [1] were known when L was a more general orthomodular lattice [7]. Some of these lattices can be represented as subsets of a Hilbert lattice [5, 6].

VIII. SPECTRAL STRUCTURE AND CANONICAL FORMS

In dimension $d=2$, all spectral properties reduce to the Bloch form of Sec. V. We analyse the general properties of the structure operator $T=\sum_{i=1}^{k} P_{a_i}$ associated with a k-tuple of unit vectors $(a_1,\ldots,a_k)$ in H.

A. Gram representation and spectral equivalence

Definition 19 (Analysis operator and Gram matrix)**.** Let $A\colon \mathbb{C}^k \to H$ be defined by $Ae_i = a_i$, where (e_i) is the standard basis of $\mathbb{C}^k$. The *analysis operator* is $A^\dagger\colon H \to \mathbb{C}^k$, and the *Gram matrix* is

$$G := A^\dagger A = \big(\langle a_i, a_j\rangle\big)_{i,j=1}^{k}.$$

Proposition 20 (Frame operator factorisation)**.** *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

Proposition 21 (Spectral equivalence)**.** *The nonzero spectra of T and G coincide (including multiplicities):*

$$\sigma(T)\setminus\{0\} = \sigma(G)\setminus\{0\}.$$

In particular, $\operatorname{rank}(T) = \operatorname{rank}(G) \le \min(k,n)$.

Proof. This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda(A^\dagger v)$ and $A^\dagger v \neq 0$. □

B. Spectral constraints

Proposition 22 (Trace identities)**.** *The structure operator satisfies*

$$\operatorname{tr}(T) = k, \qquad \operatorname{tr}(T^2) = \sum_{i,j=1}^{k} |\langle a_i, a_j\rangle|^2.$$

The diagonal terms $i=j$ each contribute 1*, giving* $\operatorname{tr}(T^2) \ge k$*, with equality if and only if the vectors are mutually orthogonal.*

Proof. $\operatorname{tr}(T) = \sum_i \operatorname{tr}(P_{a_i}) = k$. For the second identity, $\operatorname{tr}(T^2) = \sum_{i,j} \operatorname{tr}(P_{a_i}P_{a_j}) = \sum_{i,j} |\langle a_i, a_j\rangle|^2$. Diagonal terms contribute k; off-diagonal terms are non-negative, and vanish simultaneously if and only if all off-diagonal inner products are zero. □

Proposition 23 (Non-orthogonality and spectral spread)**.** *If the vectors (a_i) are not mutually orthogonal, then* $\operatorname{tr}(T^2) > k$*, the spectrum of T is non-flat, and $\lambda_{\max}(T) > 1$.*

Proposition 24 (Majorisation constraint)**.** *Suppose $k \le n$. Let $\lambda(T) = (\lambda_1,\ldots,\lambda_n)$ be the eigenvalues of T in decreasing order (padded with zeros to length n), and let $d = (1,\ldots,1,0,\ldots,0) \in \mathbb{R}^n$ have k ones and $n-k$ zeros. Then $\lambda(T) \succ d$.*

Proof. The diagonal entries of G are $G_{ii} = \|a_i\|^2 = 1$, so the diagonal of G is $(1,\ldots,1) \in \mathbb{R}^k$. By the Schur–Horn theorem, $\lambda(G) \succ (1,\ldots,1)$. Since G and T share the same nonzero eigenvalues (Proposition 21), padding $\lambda(G)$ with $n-k$ zeros gives $\lambda(T)$, and $\lambda(T) \succ d$ follows.

Overcomplete case $k > n$. When $k > n$ the Gram matrix G is $k \times k$ with rank at most n; the majorisation is then best stated for the k-dimensional eigenvalue vector of G against $(1,\ldots,1) \in \mathbb{R}^k$, and the Schur–Horn argument applies to G directly. □

Proposition 25 (Operator bounds)**.** *The structure operator satisfies $0 \le T \le kI$. Moreover, if $\mu := \max_{i\neq j} |\langle a_i, a_j\rangle|$, then*

$$\lambda_{\max}(T) \le 1 + (k-1)\mu.$$

Proof. Positivity $T \ge 0$ is immediate, and $T \le kI$ since each $P_{a_i} \le I$. For the eigenvalue bound, apply the Geršgorin circle theorem to G: since $G_{ii} = 1$ and $|G_{ij}| \le \mu$ for $i \neq j$, every eigenvalue of G lies in a disk centred at 1 with radius at most $(k-1)\mu$. Since the nonzero eigenvalues of T and G coincide, $\lambda_{\max}(T) \le 1 + (k-1)\mu$. □

C. Commutant and symmetry

Definition 26 (Commutant)**.** The *unitary commutant* of T is $\mathscr{C}(T) := \{U \in U(H) : UT = TU\}$.

Proposition 27 (Structure of the commutant)**.** *Let $T = \sum_\alpha \lambda_\alpha P_\alpha$ be the spectral decomposition of T with distinct eigenvalues λ_α and orthogonal projections P_α of ranks m_α. Then $\mathscr{C}(T) \cong \prod_\alpha U(m_\alpha)$.*

Remark 28. Spectral degeneracies ($m_\alpha > 1$) give rise to continuous families of distinct rank-one decompositions of T (as in tight frames), whereas simple spectrum leads to strong rigidity. This explains why the tight-frame case (Remark 15) is so rich in pseudocontexts.

D. Canonical normal form

Proposition 29 (Spectral normal form)**.** *Every structure operator is unitarily equivalent to* $\operatorname{diag}(\lambda_1,\ldots,\lambda_r,0,\ldots,0)$ *with $\lambda_i > 0$, $\sum_{i=1}^{r} \lambda_i = k$, and $r = \operatorname{rank}(T)$.*

E. Canonical parametrisation of all decompositions

Theorem 30 (Canonical decomposition)**.** *Let T be a positive operator of rank r with spectral decomposition $T = \sum_{i=1}^{r} \lambda_i |e_i\rangle\langle e_i|$, $\lambda_i > 0$, and set $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$.*

Every rank-one decomposition $T = \sum_{j=1}^{k} P_{a_j}$ with $\|a_j\| = 1$ corresponds to a matrix $C \in \mathbb{C}^{r\times k}$ satisfying

$$CC^\dagger = \Lambda, \qquad \|c_j\|^2 = 1 \text{ for all } j, \tag{6}$$

where c_j is the j-th column of C, via $a_j = \sum_{i=1}^{r} c_{ij} e_i$. Conversely, every such C defines a valid decomposition.

Proof. Let $T = AA^\dagger$ with columns a_j. In the eigenbasis of T, write $A = UC$ where U is the unitary diagonalising T on its support. Then $CC^\dagger = U^\dagger T U\big|_{\mathrm{supp}(T)} = \Lambda$, and $\|c_j\| = \|a_j\| = 1$. The converse is a direct computation. □

Corollary 31 (Unitary freedom)**.** *Among all rank-one decompositions of T into k (not necessarily unit-norm) vectors, any two are related by $A \mapsto AU$ for some $U \in U(k)$. Imposing the unit-norm constraint (6) cuts out a submanifold of this $U(k)$-orbit; the action of $U(k)$ is not transitive on unit-norm decompositions in general.*

F. Characterisation of pseudocontexts

Proposition 32 (Operator equivalence criterion)**.** *Two k-tuples (a_i) and (b_i) of distinct rays form a pseudocontext if and only if their matrices C_A, C_B in the eigenbasis of T both satisfy (6):*

$$C_A C_A^\dagger = C_B C_B^\dagger = \Lambda.$$

Remark 33. Pseudocontexts correspond to distinct points in the space $\{C \in \mathbb{C}^{r\times k} : CC^\dagger = \Lambda, \|c_j\| = 1\}/\sim$, where $\sim$ identifies decompositions that yield the same set of rays.

G. Extremal spectral bounds

Proposition 34 (Rayleigh bounds)**.** *For any density operator ρ,*

$$\lambda_{\min}(T) \le \mathrm{tr}(\rho T) \le \lambda_{\max}(T).$$

Both bounds are tight, achieved by pure states on the corresponding eigenvectors of T.

H. Examples of nondegenerate spectra

Explicit three-dimensional implementations of nondegenerate structure operators were derived in [5].

- **15-atom hypergraph (T_1).** A pseudocontext of size $k = 3$ from a generalised firefly logic yields

$$\lambda_1 = 2, \;\; \lambda_2 = \tfrac{7+\sqrt{21}}{14} \approx 0.827, \;\; \lambda_3 = \tfrac{7-\sqrt{21}}{14} \approx 0.173,$$

  with $\mathrm{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 23.

- **36-atom hypergraph (T_2).** A larger true-implies-false gadget gives $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ with $\mathrm{tr}(T_2) = 3$ and $\lambda_{\max} = 13/5 > 1$.

IX. PHYSICAL INTERPRETATION AND CONSTRAINTS

A pseudocontext $(\{a_i\}, \{b_i\})$ implements a state-independent empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \longmapsto \sum_i \mathrm{tr}(\rho A_i), \qquad \rho \longmapsto \sum_i \mathrm{tr}(\rho B_i)$$

agree on the whole state space.

The spectral decomposition of T establishes absolute quantum limits on these probability sums. By Proposition 34,

$$\lambda_{\min}(T) \le \sum_{i=1}^{k} P(a_i) \le \lambda_{\max}(T). \tag{7}$$

These bounds are tight.

For the 36-atom hypergraph example [5], the structure operator $T_2 = \mathrm{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued ($\{0,1\}$) truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over $\{0,1,2,3\}$, which includes the extreme values 0 and 3. Because T_2 has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extremes; the quantum system cannot reach them. This discrepancy is a signature of Kochen–Specker-type contextuality [8, 9].

This state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [10], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. A precise comparison is beyond the scope of this note.

In the language of quantum information, $\{A_i\}$ and $\{B_i\}$ are distinct rank-one decompositions of the same positive operator. When this operator is proportional to the identity on its support, the families form tight frames; after normalisation they yield POVMs.

X. OPEN PROBLEMS

Open Problem (Classification). Classify, up to unitary equivalence, all pseudocontexts of size k in $\mathbb{C}^n$. Even for small (k, n), the orbit structure is not yet transparent.

Open Problem (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [5, 6].

Open Problem (Frame-symmetry characterisation). Can one characterise pseudocontexts purely in terms of symmetries of the underlying rank-one frame, for example in terms of unitaries commuting with the structure operator and permuting the atoms of the two decompositions?

Remark 35 (Minimal ambient dimension). Every pseudocontext of size $k \geq 2$ can be realised inside a two-dimensional subspace, and hence embedded into $\mathbb{C}^n$ for any $n \geq 2$. In dimension $d = 2$ a complete classification is given in Sec. V. For $k \geq 3$ the construction also works over $\mathbb{R}^n$. No pseudocontext of any size exists in dimension $n = 1$.

Open Problem (Minimal size over $\mathbb{R}$). Theorem 10 establishes that genuine real pseudocontexts of size $k = 2$ do not exist in any dimension, while Example 16 demonstrates their existence for $k = 3$. Thus, $k = 3$ is the minimal size for genuine real pseudocontexts. The remaining open problem is to classify all such minimal ($k = 3$) examples up to orthogonal equivalence over $\mathbb{R}$.

ACKNOWLEDGMENTS

This research was funded in part by the Austrian Science Fund (FWF) 10.55776/PIN5424624 and the Czech Science Foundation (GAČR) 25-20013L. The authors declare no conflict of interest.

[1] R. Mayet, M. Navara, and V. Rogalewicz, Orthomodular lattices with rich state spaces, Algebra Universalis **43**, 1 (2000).
[2] A. M. Gleason, Measures on the closed subspaces of a Hilbert space, Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal) **6**, 885 (1957).
[3] J. M. Renes, R. Blume-Kohout, A. J. Scott, and C. M. Caves, Symmetric informationally complete quantum measurements, Journal of Mathematical Physics **45**, 2171 (2004).
[4] M. Navara and K. Svozil, Pseudocontexts, Presentation at *Quantum Structures 2024* (2024), july 1–5, 2024.
[5] M. Navara and K. Svozil, Form of contextuality predicting probabilistic equivalence between two sets of three mutually noncommuting observables, Physical Review A **109**, 022222 (2024), arXiv:2309.13091.
[6] M. Navara and K. Svozil, Exploring quantum contextuality with the quantum Möbius-Escher-Penrose hypergraph, Physical Review A **111**, 042209 (2025).
[7] G. Kalmbach, *Orthomodular Lattices*, London Mathematical Society Monographs, Vol. 18 (Academic Press, London and New York, 1983).
[8] A. Cabello, J. M. Estebaranz, and G. García-Alcaine, Bell-Kochen-Specker theorem: A proof with 18 vectors, Physics Letters A **212**, 183 (1996), arXiv:quant-ph/9706009.
[9] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics, Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal) **17**, 59 (1967).
[10] R. W. Spekkens, Contextuality for preparations, transformations, and unsharp measurements, Physical Review A **71**, 052108 (2005), arXiv:quant-ph/0406166.